

Hi, this is Monica Rajagopal, presenting Decision Packet Automation for Life Underwriting and Claims.

The scope of this proof of concept is limited to underwriting only.

Life insurers receive large volumes of unstructured information, including medical reports, APS documents, lab results, applications, policy records, claim forms, death certificates, and case notes. Today, underwriters and claims examiners often spend significant time reading documents, locating key evidence, creating summaries, checking guidelines, and documenting their decisions.

Our solution uses Claude AI, document intelligence, OCR, and a configurable business rules engine to automate the preparation of a complete, evidence-grounded Decision Packet.

The process begins when documents and case data are received. OCR converts scanned files into usable text, and Claude extracts relevant information such as diagnoses, medications, lab results, dates, policy details, claim details, and other risk indicators. From there, each case moves through four stages: ingest and extract, normalise and cite, validate, and decide.

The solution then normalises this information into a structured evidence model. Every key finding is linked back to its source document, page number, date, value, and confidence score. This creates a transparent medical or claims timeline that the reviewer can quickly validate.

Next, the tool checks whether the evidence is complete and consistent. If information is missing, conflicting, outdated, or uncertain, the system does not make an unsupported recommendation. Instead, it creates a targeted request for the exact evidence needed, and routes the case for review.

For cases with sufficient evidence, the solution retrieves carrier-approved underwriting guidelines and applies them through a controlled rules engine. Claude then creates a Decision Packet containing a case summary, relevant findings, source citations, triggered rules, required actions, and a recommended disposition.

The underwriter or claims examiner remains in control. They can approve, modify, override, request additional information, or escalate the case to a senior underwriter, a medical reviewer, the special investigations unit, or legal.

This solution helps insurers reduce manual document review, improve decision consistency, accelerate turnaround time, and create audit-ready documentation. It can begin as a focused proof of value, and scale across underwriting and claims workflows through integrations with policy, claims, document management, and data platforms.

Decision Packet Automation turns fragmented insurance documents into a clear, traceable, and reviewer-ready decision experience.

That is the solution in outline. Now let me show you the working prototype.

On screen is a single term life case, exactly as it arrives. Six documents: the application, an attending physician statement, a lab report, a motor vehicle record, the prior underwriting history, and an email from the advisor. In production these are produced by OCR from scanned PDFs, faxes and images. Here they have already been converted to text, which is where our pipeline picks up.

Notice that the formats are mixed: plain text, a spreadsheet, a PDF and a Word document. That is normal for a real case file, and the pipeline handles all of them.

Now we run the pipeline for this case with a single command.

Stage one is ingestion. Every document is opened, the text is pulled out and split into chunks, and each chunk is tagged with the case identifier and the type of document it came from. No judgement is applied here. This is purely reading files.

Stage two is extraction. The tagged text goes to Claude, which pulls out the underwriting facts: the applicant details, the diagnosis, the lab values, the driving record, the prior decision. Alongside the facts it records what is missing, any risk signals, and any place where two documents disagree with each other. Every fact is tied back to the exact sentence it came from.

Stage three is the narrative. Claude takes those structured findings and writes the reviewer-facing brief: a short summary, a set of concrete next steps, and a draft recommendation.

Stage four is scoring. Claude maps the findings onto our underwriting framework, which has nineteen factors, each with four defined risk levels. Claude only chooses which level applies. The arithmetic is then done in plain Python: each factor is multiplied by its weight, summed to a total, and matched to a risk band. That means the same findings always produce exactly the same score.

The packet is now assembled.

On disk we have four files for this case. The tagged text from stage one. The extracted facts with their citations. The risk score with its factor breakdown. And the final packet, which combines all of it.

This is the reviewer's view. Each case shows its risk band, its score, and the suggested action, colour coded so a queue can be triaged at a glance. The spread runs all the way from Low, at a score of zero, up to Very High at ninety eight.

Opening a case gives us the full packet.

At the top are the headline numbers: the risk score, the risk band, and the suggested underwriting action that the framework attaches to that band.

Below that is the summary. This is the brief a reviewer reads first.

Then the key fields. Every value carries the document it came from and the exact quoted words that support it, so the reviewer can confirm a fact without reopening the source file. Each field is also marked as either stated directly or inferred, with a confidence level.

Missing information lists what a reviewer would expect to see but cannot find in the file, and why it matters.

Risk flags are categorised and rated by severity, each with its own citation.

The score breakdown shows all nineteen factors: the band chosen, the weight, and the contribution to the total. Where the file contains no evidence for a factor, it is marked as not addressed and scored favourably, rather than the system guessing.

Then the suggested action path, which gives concrete next steps in priority order, and the draft rationale behind the recommendation.

Finally the governance block records the review status and confirms that every citation in the packet resolves back to a real source document.

Every packet is a draft prepared for an underwriter. It is not an approval, a decline, or a binding decision. The system assembles and cites the evidence. The underwriter makes the call.